Understanding Switch Security

Common Threats to Physical Installations

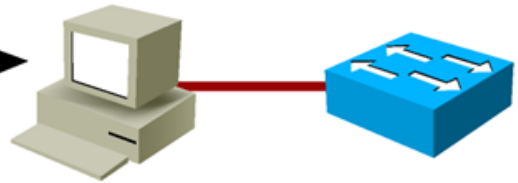
- Hardware threats
- Environmental threats
- Electrical threats
- Maintenance threats



Configuring a Switch Password

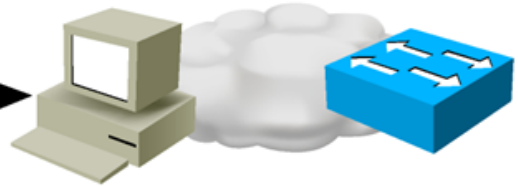
Console Password

```
SwitchX(config)#line console 0
SwitchX(config-line)#login
SwitchX(config-line)#password cisco
```



Virtual Terminal Password

```
SwitchX(config)#line vty 0 4
SwitchX(config-line)#login
SwitchX(config-line)#password sanjose
```



Enable Password

```
SwitchX(config)#enable password cisco
```



Secret Password

```
SwitchX(config)#enable secret sanfran
```

Service Password-Encryption Commands

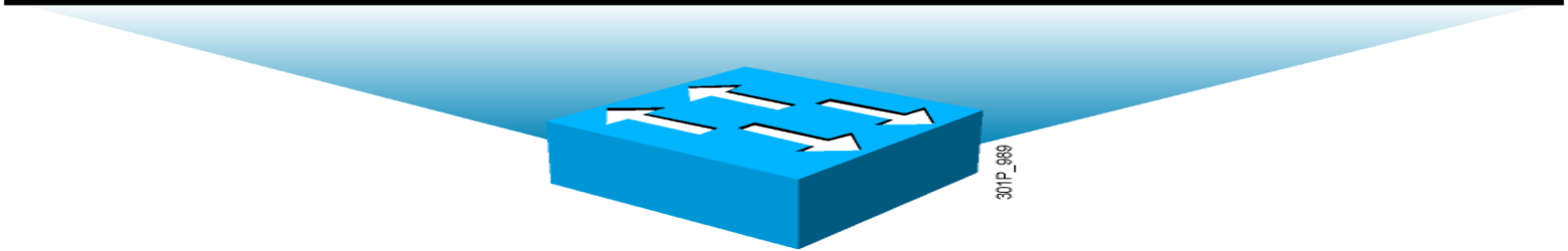
```
SwitchX(config)#service password-encryption
SwitchX(config)#no service password-encryption
```




Configuring the Login Banner

- Defines and enables a customized banner to be displayed before the username and password login prompts.

```
SwitchX# banner login " Access for authorized users only. Please enter your username and
password. "
```



Configuring Port Security

- Cisco Catalyst 2960 Series

```
SwitchX(config-if)# switchport port-security [  mac-address
mac-address | mac-address sticky [mac-address] | maximum value
| violation {restrict | shutdown}]
```

```
SwitchX(config)# interface fa0/5
SwitchX(config-if)# switchport mode access
SwitchX(config-if)# switchport port-security
SwitchX(config-if)# switchport port-security maximum 1
SwitchX(config-if)# switchport port-security mac-address sticky
SwitchX(config-if)# switchport port-security violation shutdown
```



Verifying Port Security on the Catalyst 2960 Series

```
SwitchX# show port-security [interface interface-id]
[address] [ | {begin | exclude | include} expression]
```

```
SwitchX# show port-security interface fastethernet 0/5
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 20 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address        : 0000.0000.0000
Security Violation Count   : 0
```



Verifying Port Security on the Catalyst 2960 Series (Cont.)

```
SwitchX# shor port-security address
          Secure Mac Address Table
-----------------------------------------------------------------------
Vlan    Mac Address       Type                  Ports   Remaining Age
                                                          (mins)
----    -----------       ----                  -----   -------------
 1      0008.dddd.eeee    SecureConfigured      Fa0/5         -
-----------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 1024
```

```
SwitchX# show port-security
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
               (Count)       (Count)          (Count)
-----------------------------------------------------------------------------
      Fa0/5        1             1                0          Shutdown
-----------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 1024
```

Securing Unused Ports

- Unsecured ports can create a security hole.
- A switch plugged into an unused port will be added to the network.
- Secure unused ports by disabling interfaces (ports).



Disabling an Interface (Port)

```
SwitchX(config-int)# shutdown
```

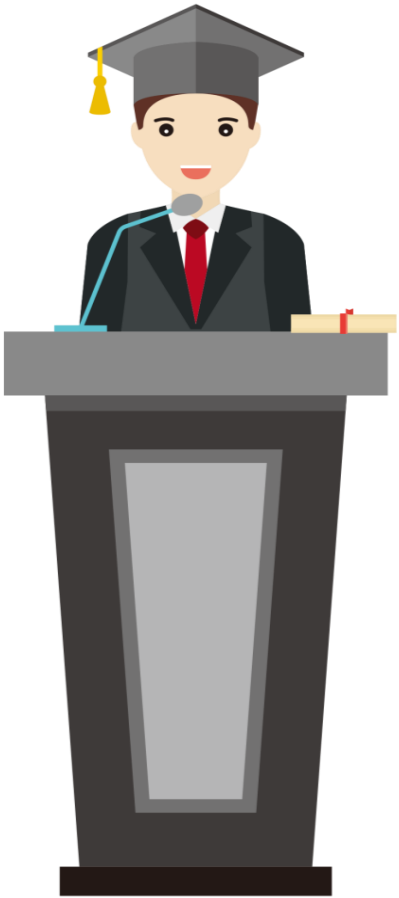
- To disable an interface, use the shutdown command in interface configuration mode.
- To restart a disabled interface, use the no form of this command.



Summary

- The first level of security is physical.
- Passwords can be used to limit access to users that have been given the password.
- The login banner can be used to display a message before the user is prompted for a username.
- Telnet sends session traffic in cleartext; SSH encrypts the session traffic.
- Port security can be used to limit MAC addresses to a port.
- Unused ports should be shut down.





THANK YOU!

